

Chapter V: Exercise 1 - Part 3

Palaeogeography Meets Palaeoclimatology

i This exercise in brief

Big Question

How did Earth's past climate vary across geological time, and how can we reconstruct it from numerical model outputs?

What You Will Do

In this module, you will:

1. Load and inspect a NetCDF climate file using `xarray`
2. Understand the structure and metadata of NetCDF files
3. Convert raw model outputs to standard units
4. Classify every grid cell into a Köppen-Geiger climate zone
5. Fetch a land mask from palaeogeography and apply it to your map

Why This Matters

Climate models produce outputs in formats that are very different from the GeoTIFF rasters you have worked with so far. NetCDF is the standard format in climate and ocean sciences, and knowing how to read, inspect, and analyse these files is an essential skill for any geoscientist working with model data. Köppen-Geiger classification is a widely used framework to summarise complex climate information into a simple, interpretable map.

Your Tasks

1. **Load the data:** Open a NetCDF file for a chosen reconstruction age.
 - 1.1. Inspect the dimensions, coordinates, and variables.
 - 1.2. Read the variable attributes to understand units and provenance.

2. **Explore the data:** Select and print subsets of the data.
 - 2.1. Extract temperature values for a single month.
 - 2.2. Extract a time series for a specific location (e.g. Geneva).
3. **Convert units:** Transform raw model outputs into usable units.
 - 3.1. Convert temperature from Kelvin to Celsius.
 - 3.2. Convert precipitation from m s^{-1} to mm/month.
4. **Classify climate zones:** Apply the Köppen-Geiger algorithm.
 - 4.1. Compute temperature and precipitation derived quantities.
 - 4.2. Write and apply the classification function grid cell by grid cell.
5. **Mask and visualise:** Fetch a land mask and plot the final map.
 - 5.1. Request the palaeogeography layer from the PANALEGIS GeoServer.
 - 5.2. Resample the mask to match the climate grid resolution.
 - 5.3. Apply the mask and plot the Köppen-Geiger map.

Skills You Will Gain

1. Reading and inspecting NetCDF files with `xarray`.
2. Understanding CF Conventions and variable metadata.
3. Performing unit conversions on multi-dimensional arrays.
4. Implementing a rule-based classification algorithm in Python.
5. Combining data from different sources and resolutions using resampling.
6. Fetching raster data from a WCS GeoServer endpoint.

Points for Discussion

The climate grid has a resolution of $\sim 5.6^\circ$. How does this affect the reliability of the Köppen-Geiger classification, particularly near coastlines or mountain ranges?

The land mask is derived from elevation data ($0 \text{ m} = \text{land}$). What are the limitations of this approach for defining land vs. ocean?

We used a simplified aridity threshold for group B climates. How might results differ if you used the full seasonality-dependent formula?

How would you expect the Köppen-Geiger map to change between 0 Ma and 100 Ma?

What geological or astronomical factors might explain the differences?

Step-by-step instructions

In the second part of the exercise, we extended our analysis on other maps beyond palaeogeography:

- Defined functions to easily request a WCS for a given layer name and reconstruction age, and plot the map
- Explored options to analyze the map, using `rasterio.plot.show_hist` or its mean value.
- Optimized our analysis by using `xarray` and `rioxarray` to work efficiently with several GeoTIFF layers that share the same extent and resolution.
- Performed operations to filter data between layers to keep only pixels of interest using `numpy`
- Explored the relationship between the oceanic lithosphere and the seafloor age.
- Extended our analysis to the entire Phanerozoic for continental and oceanic domains.
- Exported our results in tabular format using `pandas`.

In this last part, we will have a look at another data format that is very common in the geosciences: NetCDF (abbreviation for Network Common Data Form). Unlike the GeoTIFF format that requires one file per map, a NetCDF is like a labelled can hold many variables (temperature, precipitation...) each with their own dimensions (space, time) and metadata, all in one file.

Feature	GeoTIFF	NetCDF
File extension	<code>.tif</code>	<code>.nc</code>
Typical use	Single raster layer (e.g. one image)	Multiple variables + time dimensions
Dimensions	2D (rows $\times$ cols)	N-dimensional (e.g. month $\times$ lat $\times$ lon)
Metadata	Limited	Rich, self-describing
Common in	Remote sensing, GIS	Climate science, oceanography
Python library	<code>rasterio</code> , <code>rioxarray</code>	<code>xarray</code> , <code>netCDF4</code>

We will learn how to read NetCDF files, understand their metadata, and turn input data related to precipitation and temperature into climate zones maps (Köppen-Geiger) for the entire Phanerozoic.

Let's first load a file from the `data/climate` folder. We have one file per reconstruction age. The last 4 digits in the filename represents the age in millions of years (Ma), written to 4 digits style. For example, 0 Ma $\rightarrow$ 0000, 100 Ma $\rightarrow$ 0100, 444 Ma $\rightarrow$ 0444. By using a regular expression (regex) from the `re` library, selecting an age can be done like:

```

import xarray as xr
import numpy as np
age = 100

nc_path = f"data/climate/panalesis_version_0_plasim-genie_tas_pr_{str(age).zfill(4)}Ma.nc"

ds = xr.open_dataset(
    nc_path,
    engine="netcdf4"
)
print(ds)

```

```

<xarray.Dataset> Size: 197kB
Dimensions:                (month: 12, latitude: 32, longitude: 64)
Coordinates:
  * month                   (month) int32 48B 1 2 3 4 5 6 7 8 9 10 11 12
  * latitude                (latitude) float32 128B 85.76 80.27 ... -85.76
  * longitude               (longitude) float64 512B -180.0 -174.4 ... 174.4
Data variables:
  temperature_surface_air   (month, latitude, longitude) float32 98kB ...
  precipitation_total       (month, latitude, longitude) float32 98kB ...
Attributes:
  title:                    Monthly mean near-surface air temperature and tota...
  institution:              ETH Zürich - Dept. of Earth & Planetary Sci. © Nik...
  Conventions:              CF-1.8
  pCO2:                     1947.002
  solar_constant:           1353.5
  pCO2_units:               ppm
  solar_constant_units:     W m-2
  age:                      100.0
  age_units:                Ma (millions of years before present)
  paleogeography:           Panalesis version 0 (Franziskakis et al. 2025)
  length_of_simulation:     3000 model years with 1:10 atmosphere:ocean gearing

```

→ Have a look at the metadata: how is the data organized ? What is the resolution ?

→ Does this give you enough information about the data source ?

Every variable in a NetCDF file carries **attributes**: metadata describing what the variable is, its units, and more. This follows the **CF Conventions** standard. Let's have a look at the air surface temperature variable:

```
print(ds["temperature_surface_air"].attrs)
```

```
{'standard_name': 'air_temperature', 'long_name': 'Near-surface air temperature', 'units': 'K'}
```

Thanks to the structure of the input data and how we read it using `xarray`, it is possible to access only selected slices of the entire dataset. For instance, we can select only one variable:

```
temp = ds["temperature_surface_air"]
print(temp)
```

```
<xarray.DataArray 'temperature_surface_air' (month: 12, latitude: 32,
                                             longitude: 64)> Size: 98kB
[24576 values with dtype=float32]
Coordinates:
  * month      (month) int32 48B 1 2 3 4 5 6 7 8 9 10 11 12
  * latitude   (latitude) float32 128B 85.76 80.27 74.74 ... -80.27 -85.76
  * longitude  (longitude) float64 512B -180.0 -174.4 -168.8 ... 168.8 174.4
Attributes:
  standard_name:  air_temperature
  long_name:      Near-surface air temperature
  units:          K
```

We can also select a single month, using `.sel()` which selects data by label (e.g. `month=1`):

```
temp_jan = temp.sel(month=1)
print(temp_jan)
```

```
<xarray.DataArray 'temperature_surface_air' (latitude: 32, longitude: 64)> Size: 8kB
[2048 values with dtype=float32]
Coordinates:
  * latitude   (latitude) float32 128B 85.76 80.27 74.74 ... -80.27 -85.76
  * longitude  (longitude) float64 512B -180.0 -174.4 -168.8 ... 168.8 174.4
  month       (month) int32 4B 1
Attributes:
  standard_name:  air_temperature
  long_name:      Near-surface air temperature
  units:          K
```

Or even a given location. For instance, if we take the approximate location of Geneva (lat/lon). The `method="nearest"` finds the grid cell with the centroid that is closest to your input coordinates:

```
temp_geneva = temp.sel(latitude=46.2, longitude=6.1, method="nearest")
print(temp_geneva)
print(temp_geneva.values)
```

```
<xarray.DataArray 'temperature_surface_air' (month: 12)> Size: 48B
[12 values with dtype=float32]
Coordinates:
  * month      (month) int32 48B 1 2 3 4 5 6 7 8 9 10 11 12
    latitude   float32 4B 47.07
    longitude  float64 8B 5.625
Attributes:
  standard_name:  air_temperature
  long_name:      Near-surface air temperature
  units:          K
[282.61826 282.5746 283.89636 287.60983 290.13657 293.08612 295.2669
 295.68762 293.59634 289.4877 285.94287 283.80334]
```

→ Notice how the basic print returns the entire subset structure with the new dimensions and attributes. If you want to print the values, you need to specify it, but ****be careful** to do this only when you have a small subset.

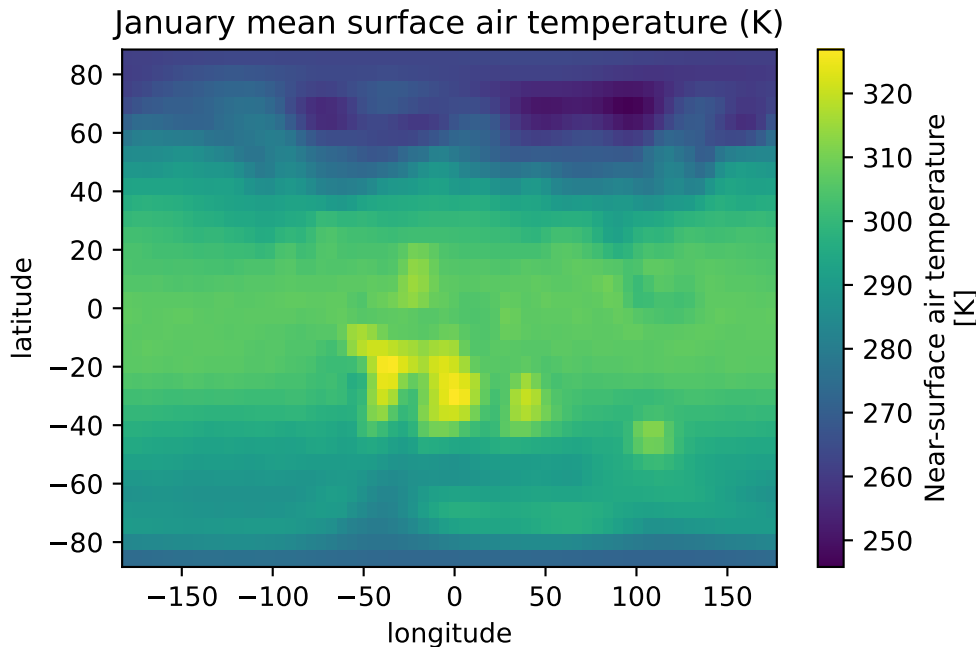
→ We asked for the Geneva location, but where is the closest pixel centroid point located ?

→ What were the temperatures 100 millions years ago at this location?

→ What does this tell you about the model resolution ?

Let's now have a look at our data by plotting it, reusing the January subset we did before.

```
import matplotlib.pyplot as plt
temp_jan.plot()
plt.title("January mean surface air temperature (K)")
plt.show()
```



As you noticed, the units for temperature are in Kelvin and precipitation are in meters per seconds. Before any analysis, we need to convert both variables to standard units, in our case this would be degrees Celsius and millimeters per month.

```
temp_c = ds["temperature_surface_air"] - 273.15
temp_c.attrs["units"] = "°C"

seconds_per_day = 86400
days_per_month = [31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31]
days = xr.DataArray(days_per_month, coords=[ds.month], dims=["month"])

precip_mm = ds["precipitation_total"] * seconds_per_day * days * 1000
precip_mm.attrs["units"] = "mm/month"
```

We now have everything we need to classify each grid cell into a climate zone.

For the classification we need to know how warm or cold each grid cell gets. The coldest and warmest month determine whether a climate is polar, temperate, or continental. The number of months above 10°C is a proxy for the growing season.

```
t_annual = temp_c.mean(dim="month")
t_min = temp_c.min(dim="month")
t_max = temp_c.max(dim="month")
months_above_10 = (temp_c > 10).sum(dim="month")
```

Precipitation seasonality is just as important as the total amount. A climate with dry summers behaves very differently from one with dry winters, even if the annual total is the same. We split the year into summer (Apr–Sep) and winter (Oct–Mar) halves to detect this seasonality.

```
p_annual = precip_mm.sum(dim="month")
p_min = precip_mm.min(dim="month")
p_max = precip_mm.max(dim="month")
p_summer = precip_mm.sel(month=[4,5,6,7,8,9]).sum(dim="month")
p_winter = precip_mm.sel(month=[1,2,3,10,11,12]).sum(dim="month")
```

Group B (arid) climates are defined by precipitation being too low relative to evaporation demand. Evaporation increases with temperature, so the threshold is not a fixed number but depends on the annual mean temperature. We will use a simplified version of the standard formula:

```
p_threshold = 2 * t_annual + 28
```

The full Köppen formula adjusts `p_threshold` based on precipitation seasonality (adding 0 to 28 mm depending on whether rain falls in summer or winter). Here we use a fixed offset of +28 for simplicity.

Let's write a function that takes the pre-computed quantities for a single grid cell and returns its 2-letter Köppen-Geiger code. The logic follows the standard hierarchical order: polar first, then arid, tropical, temperate and continental.

```
def classify_koppen(t_ann, t_mn, t_mx, p_ann, p_mn, p_sum, p_win, p_thr):

    # E: Polar (warmest month too cold for tree growth)
    if t_mx < 0:
        return "EF"    # ice cap
    if t_mx < 10:
        return "ET"    # tundra

    # B: Arid (precipitation too low relative to evaporation)
    if p_ann < p_thr:
        if p_ann < 0.5 * p_thr:
            return "BW"    # desert
        return "BS"    # steppe

    # A: Tropical (coldest month still warm (>= 18°C))
```

```

if t_mn >= 18:
    if p_mn >= 60:
        return "Af"    # rainforest
    if p_mn >= 100 - p_ann / 25:
        return "Am"    # monsoon
    return "Aw"        # savanna

# C: Temperate (coldest month between -3°C and 18°C)
if t_mn > -3:
    if p_sum > 10 * p_win:
        return "Cs"    # dry summer
    if p_win > 3 * p_sum:
        return "Cw"    # dry winter
    return "Cf"        # no dry season

# D: Continental (coldest month below -3°C)
if p_sum > 10 * p_win:
    return "Ds"        # dry summer
if p_win > 3 * p_sum:
    return "Dw"        # dry winter
return "Df"           # no dry season

```

We now loop over every grid cell and apply the function. This will give us a numpy array (2D) with the two-letters classes for each cell.

```

n_lat = len(ds.latitude)
n_lon = len(ds.longitude)
koppen = np.full((n_lat, n_lon), "", dtype=object)

for i in range(n_lat):
    for j in range(n_lon):
        koppen[i, j] = classify_koppen(
            t_ann = float(t_annual[i, j]),
            t_mn  = float(t_min[i, j]),
            t_mx  = float(t_max[i, j]),
            p_ann = float(p_annual[i, j]),
            p_mn  = float(p_min[i, j]),
            p_sum = float(p_summer[i, j]),
            p_win = float(p_winter[i, j]),
            p_thr = float(p_threshold[i, j])
        )

```

```
print(np.unique(koppen))
```

```
['Af' 'Am' 'Aw' 'BS' 'BW' 'Cf' 'Cw' 'Df' 'Dw' 'EF' 'ET']
```

Let's now plot our map to inspect the results.

```
koppen_classes = {
    "ET": (1, "#C0C0C0"),
    "EF": (2, "#FFFFFF"),
    "Dw": (3, "#8B0000"),
    "Ds": (4, "#B22222"),
    "Df": (5, "#CD5C5C"),
    "Cw": (6, "#2E8B57"),
    "Cs": (7, "#87CEFA"),
    "Cf": (8, "#1E90FF"),
    "BW": (9, "#DAA520"),
    "BS": (10, "#F4A460"),
    "Aw": (11, "#9ACD32"),
    "Am": (12, "#32CD32"),
    "Af": (13, "#006400"),
}

koppen_num = np.zeros((n_lat, n_lon))
for i in range(n_lat):
    for j in range(n_lon):
        if koppen[i, j] in koppen_classes:
            koppen_num[i, j] = koppen_classes[koppen[i, j]][0]

from matplotlib.colors import ListedColormap, BoundaryNorm

codes = list(koppen_classes.keys())
colors = [koppen_classes[k][1] for k in codes]
cmap = ListedColormap(colors)
norm = BoundaryNorm(boundaries=np.arange(0.5, len(codes) + 1.5), ncolors=len(codes))

# Plot
fig, ax = plt.subplots(figsize=(8, 4))

im = ax.pcolormesh(
    ds.longitude.values,
    ds.latitude.values,
    koppen_num,
```

```

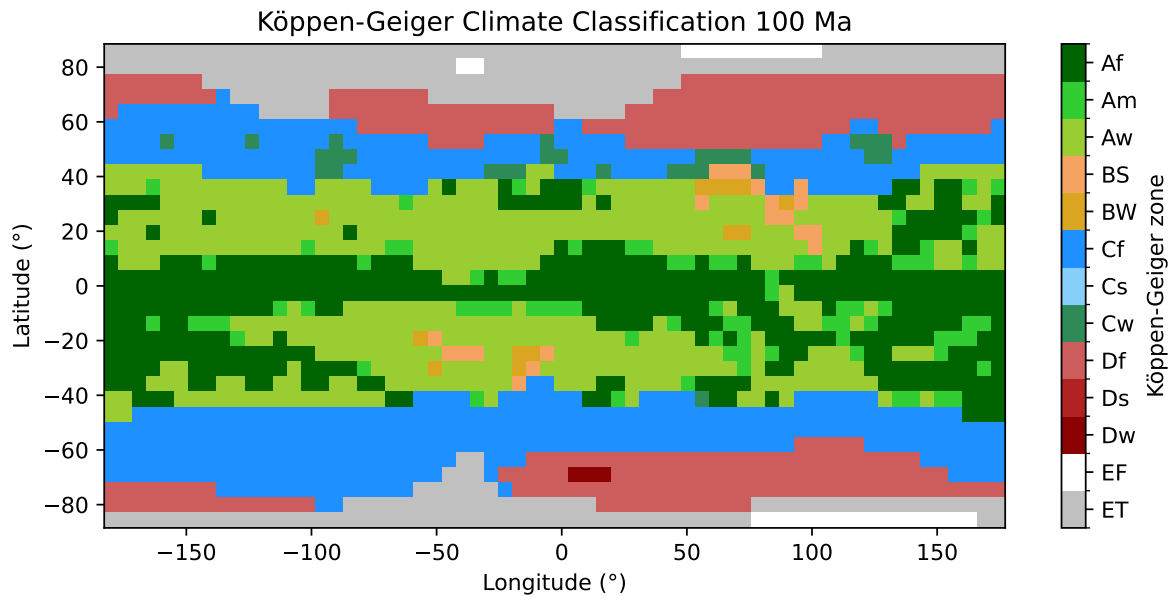
cmap=cmap,
norm=norm
)

cbar = plt.colorbar(im, ax=ax, ticks=np.arange(1, len(codes) + 1))
cbar.ax.set_yticklabels(codes)
cbar.set_label("Köppen-Geiger zone")

ax.set_title(f"Köppen-Geiger Climate Classification {int(ds.attrs['age'])} Ma")
ax.set_xlabel("Longitude (°)")
ax.set_ylabel("Latitude (°)")

plt.tight_layout()
plt.show()

```



For clarity, here is a quick overview of the 13 climate zones we classify:

Code	Name	Key characteristics
Af	Tropical rainforest	No dry season, every month > 60 mm
Am	Tropical monsoon	Short dry season, high annual rainfall
Aw	Tropical savanna	Pronounced dry winter season
BS	Semi-arid steppe	Low rainfall, not quite desert
BW	Arid desert	Very low rainfall, hot or cold
Cf	Temperate, no dry season	Mild winters, rain year-round

Code	Name	Key characteristics
Cs	Temperate, dry summer	Mediterranean-type climate
Cw	Temperate, dry winter	Mild winters, summer rains
Df	Continental, no dry season	Cold winters, rain year-round
Ds	Continental, dry summer	Cold winters, dry summers
Dw	Continental, dry winter	Very cold winters, summer rains
ET	Tundra	Warmest month between 0–10°C
EF	Ice cap	Warmest month below 0°C, permanent ice

The climate model runs over the full grid including ocean cells. To show only land areas in our Köppen-Geiger map, we will create a land mask directly from the PANALEISIS GeoServer, as we used in the first two parts of this exercise.

The time convention is the same as before: geological age in Ma is added to the year 2000 (e.g. 100 Ma → 2100-01-01T00:00:00.000Z).

```
import requests
from rasterio.io import MemoryFile
from scipy.ndimage import zoom

base_url = "https://geoserver.panaleisis.org/geoserver/"
workspace = "panaleisis_atlas_epsg_4326"
layer_name = "palaeogeography_4326"
crs = "EPSG:4326"
bbox = "-180,-90,180,90"
resx = "0.1"
resy = "0.1"
time = f"{2000 + age}-01-01T00:00:00.000Z"

wcs_url = (
    f"{base_url}{workspace}/wcs?"
    f"service=WCS&version=1.0.0&request=GetCoverage&"
    f"coverage={workspace}:{layer_name}&"
    f"crs={crs}&bbox={bbox}&resx={resx}&resy={resy}&"
    f"time={time}&format=GEOTIFF"
)

data = requests.get(wcs_url).content
with MemoryFile(data) as memfile:
    with memfile.open() as dataset:
        mask_raw = dataset.read(1).astype(float)
        nodata = dataset.nodata
```

```

    if nodata is not None:
        mask_raw[mask_raw == nodata] = np.nan

land_mask_hires = mask_raw >= 0

```

Our elevation data (from palaeogeography) has a resolution of 0.1° , which is much more higher resolution than the climate data (approx 5.6°). We need to resample it down using `scipy.ndimage.zoom` with `order=0` (nearest neighbour) and keep a cell as land if more than 50% of the high-resolution pixels are land. This is an approximation but will give us an overview globally.

```

zoom_factors = (n_lat / land_mask_hires.shape[0],
                n_lon / land_mask_hires.shape[1])
land_mask = zoom(land_mask_hires.astype(float), zoom_factors, order=0) > 0.5

print(f"Mask shape: {land_mask.shape}")

```

Mask shape: (32, 64)

→ Does the land mask shape matches the climate data ?

→ What other method could we use to have matching resolutions accross both maps?

Now that we have our land mask, we need to apply it to our Köppen-Geiger classes, by converting all pixels not on land as `np.nan`. This implies converting the original data type (integers) to float numbers, because integers cannot describe `np.nan`.

```

koppen_num_masked = koppen_num.astype(float).copy()
koppen_num_masked[~land_mask] = np.nan

```

Finally, we can plot the updated map:

```

fig, ax = plt.subplots(figsize=(8, 4))

im = ax.pcolormesh(
    ds.longitude.values,
    ds.latitude.values,
    koppen_num_masked,
    cmap=cmap,
    norm=norm
)

```

```

cbar = plt.colorbar(im, ax=ax, ticks=np.arange(1, len(codes) + 1))
cbar.ax.set_yticklabels(codes)
cbar.set_label("Köppen-Geiger zone")

ax.set_title(f"Köppen-Geiger Climate Classification {int(ds.attrs['age'])} Ma")
ax.set_xlabel("Longitude (°)")
ax.set_ylabel("Latitude (°)")

plt.tight_layout()
plt.show()

```

